Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	**Name**	Saarang Gaggar	**Email**	saarang.gaggar@tno.nl
Institute Name	Netherlands Organization for Applied Scientific Research (TNO)			
Address	Automotive Campus 30, 5708 JZ, Helmond, The Netherlands			

2. Technology Overview

Title of Proposed Technology	**Intelligent tire based real-time tire wear, road state and friction estimation for improving (automated) vehicle performance in safety critical situations and extending ODD**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical and Materials		Mechanical and Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Automobile / Rail Vehicles	Safety Improvement Technologies	Automobile / Rail Vehicles	Vehicle Intelligence Technologies
Concept of Proposed Technology	**Context and Need** Tire-road friction is one of the most important parameters governing the available grip, thereby governing the (maximum) acceleration, braking and steering capabilities of the vehicle at any time. While the friction limit is generally not a constraint in most everyday driving since that is performed well below this limit, this threshold becomes fundamental during safety-critical emergency maneuvers, such as emergency braking or steering, or while driving in adverse weather conditions. Therefore, the knowledge of real-time tire-road friction can help improve vehicle performance, and therefore, safety in such emergency situations. In case of current driver support features up to and including SAE L2 of driving automation, the driver is the end-responsible for ensuring that the vehicle is operating safely. Therefore, if the driver notices that the road conditions are slippery, then the driver intervenes, for example by reducing the speed of the vehicle, or increasing following distance to the vehicle in front, to ensure safe operation. Similarly, systems like automatic emergency braking, might only mitigate the impact of a collision instead of avoiding it when the driver doesn't respond and the tire-road friction is lower than normal. However, in the case of conditional (L3) and high (L4) automation, it is the automated vehicle that is			

completely responsible for ensuring safe operation and transferring control to the (remote) human, where necessary. Therefore, in absence of a way of ascertaining the current available maximum tire-road friction, such vehicles are restricted to a limited Operational Design Domain (ODD) – primarily driving in limited areas, and under good, dry weather conditions.

To extend the ODD (and therefore, the availability of these systems), it would be important to have a real-time, or as a next step, anticipatory estimate of tire-road friction. However, this is far from trivial. Friction is highly dependent on, amongst other things, the state of wear of the road, water/snow/ice layer depth, road design, state of wear of the tire, tread depth, tire type and more. Additionally, friction can vary over very short distances (every few meters). This makes such estimations a very complex topic that is a topic of active research.

Naturally, as the only contact point between the vehicle and the road, tires have an important role to play in estimating the real-time tire-road friction. In this regard, intelligent tire sensors can provide a next frontier of information to derive such tire-road friction estimates. Intelligent tires refer to a set of tires that include a set of sensors including, for instance, accelerometers, temperature and pressure gauges, as well as strain sensors, to estimate the current state of the tire, estimate the state of the road and gain insight on the contact patch between the tire and the road.

These tires face their own challenges –

- On the hardware side, e.g.:
  - The very harsh conditions encountered by these sensors, including the high accelerations, pressure and temperature makes the operating conditions especially challenging.
  - Powering the sensor becomes quite challenging, given the required small and flexible form factor. Furthermore, regularly replacing batteries would be quite challenging and undesired.
  - Low-latency communication of the sensor to ADAS/ADS systems that need to make decisions based on the output of the sensors.
- On the algorithmic side, e.g.:
  - Interpreting input signals, filtering and identifying the various parameters of interest.
  - Estimating the road state.
  - Estimating the tire-state.
  - Estimating the tire-road friction.
- On connection with ADAS/ADS:
  - Definition of requirements and specifications for improving the ODD of ADAS/ADS functions.
  - Interfacing and specifying this information for the ADAS/ADS.
- On validation, e.g.:
  - Requires a very high amount of data, and controlled experimentation.
  - Requires specialized test facilities.

Therefore, this technology requires active development towards both dimensions, the development of hardware, as well as algorithms.

Objectives of this project

This project aims to make a step towards increasing the maturity of such intelligent tire features and algorithms, towards real-world application. In order to do so, the project aims to:

- Define requirements based on the safety case, and the ODD.
- Define a scalable architecture that utilizes estimations of road state, and tire state to generate friction estimations. Conduct real-world validation of algorithms.

Uses of this technology

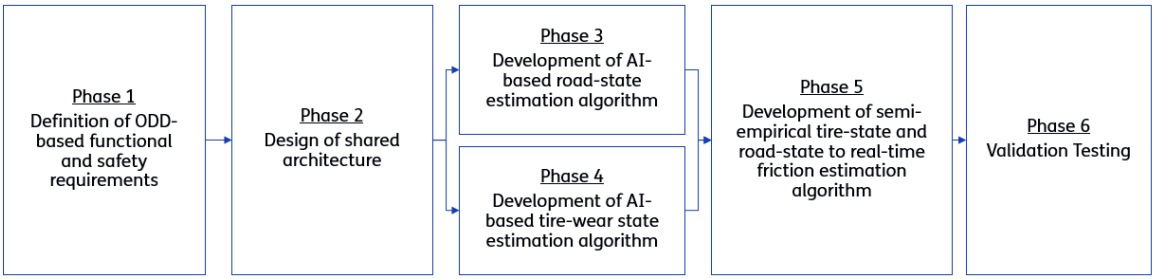
This technology will be the foundation for several applications:

1. Providing road state alerts to drivers of ADAS equipped vehicles, such that they can take intervening actions.
2. Providing alerts to the road maintenance authority, for e.g., implementing reduced speed limits or taking de-icing measures.
3. Help optimize the performance for safety-critical AD(AS) applications (e.g., ABS, AEB, etc.).
4. Contribute towards extending the ODD of conditional or highly (L3/L4) automated vehicles.

Rationale for Joint R&D Project	Real-time tire–road friction estimation is a technically complex challenge that requires the integration of advanced sensing hardware, vehicle dynamics expertise, estimation methods, and rigorous real-world validation. This multidisciplinary nature of the challenge necessitates cooperation across organizations. **Technical complexity** Accurately estimating friction in real time requires the interpretation of high-frequency signals (e.g., accelerometer, pressure, temperature) from intelligent tire sensors. These signals must be linked with rapidly changing road conditions (e.g., water, ice, snow, road wear) and tire conditions (tread depth, temperature, compound). Developing algorithms capable of inferring meaningful road-state and tire-state parameters from noisy sensor signals and translating them into reliable friction estimates requires a multidisciplinary approach spanning AI, signal processing, physics-based and semi-empirical modeling, and vehicle dynamics. **Complementarity of expertise** TNO brings decades of experience in tire modelling, vehicle dynamics, and intelligent-tire research, including the development of MF-Tyre/MF-Swift and recent work on intelligent tire-enabled vehicle safety functions. TNO also provides strong methodological capability in safety requirement specification, architecture design, and physics-informed modeling, ensuring that friction estimation is accurate, safe, and suitable for integration into ADAS/ADS systems. The Korean SME complements these strengths with its expertise in intelligent tire hardware, embedded sensing, and AI-based signal interpretation, offering capabilities that are essential for advancing state-of-the-art tire-embedded sensors and scalable algorithms. **Why international collaboration is needed** Friction estimation technologies must operate reliably across diverse climatic and road conditions. Collaborating internationally allows for access to a broader range of test environments, diverse datasets, and different tire/road types, improving model robustness. Moreover, intelligent tire solutions need alignment with global automotive markets, requiring joint development between European research organizations and Korean SMEs that supply sensor technology and target international OEMs. **Expected added value** Together, the partners can jointly create an end-to-end friction estimation solution, combining sensing hardware, AI algorithms, advanced models, safety requirements, and real-world validation. This collaboration accelerates technology maturation, improves international market readiness, and lays the foundation for future commercialization in automated vehicle platforms.
Key Words	1. Intelligent tires. 2. Real-time tire-road friction estimation. 3. Extension of ODD for automated vehicles. 4. Vehicle dynamics research.

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (2)	**End of the Project**	Level (4)
R&D Objectives	This project aims to mature and advance intelligent tire based friction estimation technologies towards a higher TRL. In order to do so, it aims to: 1. Definition of (safety-based) requirements. 2. Development of a shared architecture. 3. Develop and mature an intelligent tire-based road state estimation technology. 4. Develop and mature an intelligent tire-based tire-wear state estimation technology. 5. Development of an algorithm that translates road state estimation and tire-wear state estimation to a real-time friction estimation.			

	6. Real-world validation.
R&D Target Specifications	**AI-based road-state estimation algorithm** • **Availability:** ≥ 99% • **Classification accuracy:** ≥ 95% for major road-state classes (dry/wet/ice/snow) • **Robustness:** Performs across temperature ranges, tire types, and water/snow depth variations **AI-based tire-wear state estimation algorithm** • **Availability:** ≥ 99% • **Tread-depth estimation accuracy:** ≤ ±1.0 mm error • **Wear-classification accuracy:** ≥ 95% (new / mid-life / worn categories) **Tire–road friction estimation algorithm** • **Correlation with reference measurements:** ≥ 99% on test tracks • **Friction estimation error (μ):** ≤ ±0.1 under controlled test conditions
R&D Plan	**TNO Proposal: Intelligent tire based real-time tire wear, road state and friction estimation for improving (automated) vehicle performance in safety critical situations and extending ODD**  **Phase 1: Definition of ODD-based functional and safety requirements (Months: 0-3; Lead: TNO)** • Starting from the ODD, derive the functional and safety requirements in terms of target accuracy, availability etc. for the tire-road friction system. • Decompose these to requirements on tire-state estimation function, road-state estimation, and tire-road state estimation function. • Define a specification sheet, and validation plan. Milestone M1: Specification sheet defined. **Phase 2: Design of shared architecture (Months 3-5; Lead: TNO)** • Define an architecture and interfaces for the real-time friction estimation functionality. Milestone M2: Shared architecture defined. **Phase 3: Development of AI-based road-state estimation algorithm (Months: 5-12; Lead:**

SME)

- Conduct testing to collect intelligent tire data on various road states (e.g., wet, dry, ice, different depths of water layer etc.)
- Develop and train an AI algorithm for detecting the state of the road, based on tire sensor acceleration/pressure/temperature data.
- Test the algorithms on a test dataset , and output confidence metrics.

Milestone M3: AI-based road state estimation algorithm developed and delivered, with performance metrics per the target specification.

Phase 4: Development of AI-based tire-wear state estimation algorithm (Months: 5-12; Lead: SME)

- Conduct testing to collect intelligent tire sensor data on various states of tire wear/tread depth (e.g., summer/winter tire, new/medium/worn).
- Develop and train an AI algorithm for detecting the state of the tire, based on tire sensor acceleration/pressure/temperature data.
- Test the algorithms on a test dataset and output confidence metrics.

Milestone M4: AI-based tire state estimation algorithm developed and delivered, with performance metrics per the target specification.

Phase 5: Development of semi-empirical tire-state and road-state to real-time friction estimation algorithm (Months: 5-20; Lead: TNO)

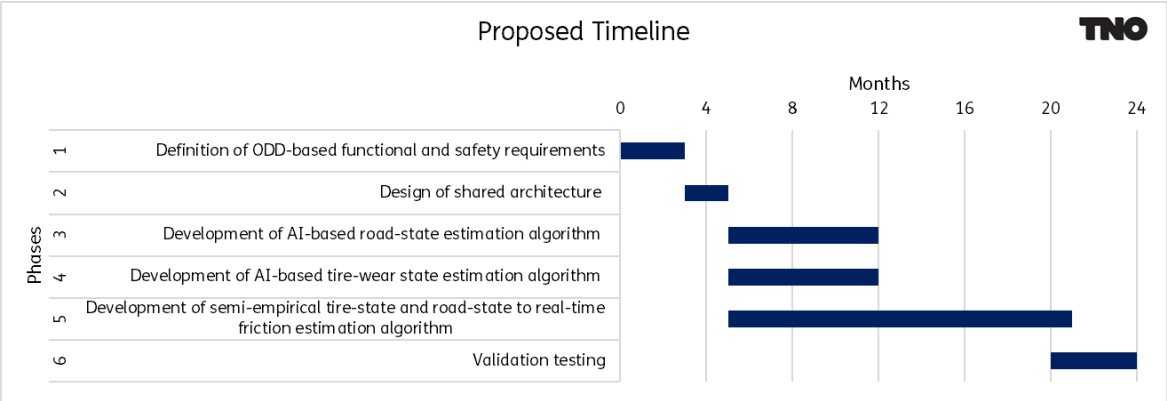
- Develop a physics-driven semi-empirical model to estimate the real-time tire-road friction estimation based on the state of the road, and state of wear of the tire.

Milestone M5: Physics-based real-time tire-road friction estimation algorithm developed.

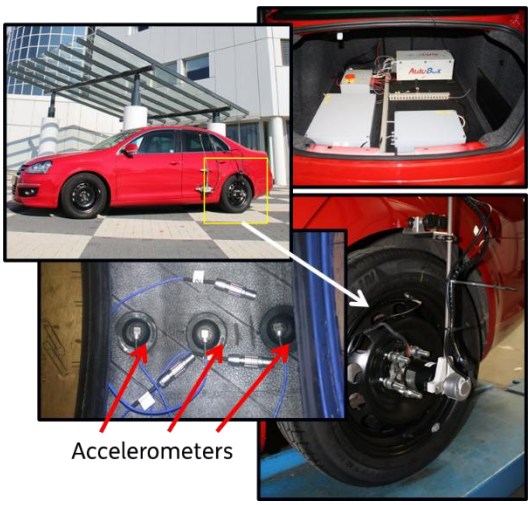
Phase 6: Validation testing (Months: 20-24; Lead: SME)

- Perform experiments to obtain tire-road friction reference values
- Perform friction estimates using intelligent tire technology
- TNO analyses the collected data and publishes a paper/report on this.

Milestone M6: Validation testing data and results. Performance analysis report by TNO.



4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**TNO's Tire Research Experience** TNO has over 30 years of prior experience with tire-oriented research. This includes work on modeling tire state [1][2][3], conducting tire testing, development of intelligent tire functionalities, and integration with vehicle ADAS/ADS solutions [4][5]. Notable examples include: • The development of the Delft-Tyre models (MF-Tyre, MF-Swift) [6] – eventually spun off by TNO in 2014 to TASS and later acquired by Siemens. This model was used by ~70% of OEMs and tire companies at the time. • Development and testing of prototype intelligent tire sensors and functionalities, as early as 2010. • Partnership with Goodyear since 2021, to utilize information about the state of the tire, to optimize vehicle safety functionalities, such as ABS and AEB. This resulted in up to a 1.5m improvement in braking performance, as showcased at the CES in 2024 and 2025.  *Example of TNO intelligent tire development and validation, in 2010.* **TNO's role in the project** TNO brings this strong background in tire-based research and vehicle safety/ADAS/ADS experience in this project. Here, TNO leads the work on the specification of requirements, development of the shared architecture and development of friction estimation algorithms. • TNO defines the safety case, and requirements specification, based on ODD and vehicle dynamics knowledge. • Development of models to estimate real-time friction estimation. • Analysis and scientific validation.
Expected R&R of Korean SME Partner	The Korean SME will be responsible for: • The Korean SME is expected to bring in expertise on intelligent tire based sensors. • Development of AI algorithms for tire-wear and road-state estimation. • Testing capabilities in lab facilities (e.g., rolling drum, or vehicle based).

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical outcomes** • A validated intelligent tire–based real-time friction estimation system combining road-state estimation, tire-state estimation, and friction prediction. • A shared reference architecture integrating intelligent tire data with vehicle ADAS/ADS decision-making. • AI models for road-state identification and tire-wear estimation, trained on real-world datasets and benchmarked against reference measurements. • A semi-empirical, physics-informed friction model linking tire and road characteristics to friction potential. • A validation dataset, test protocol, and performance analysis enabling future TRL progression. **Industrial & economic impact** • Improved safety performance of ADAS/ADS systems by enabling anticipatory control based on friction awareness (e.g., enhanced braking, steering, emergency maneuvers). • Expanded Operational Design Domain (ODD) for automated driving systems through more reliable (in-)operation in wet, icy, or worn-surface conditions. • Competitive advantage for Korean SMEs producing advanced intelligent tire sensors, enabling new product lines for global OEMs. • Foundation for commercial deployment of intelligent tire technology in premium and fleet vehicles.

7. References

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